



Mth632 final term solution made by jathol

operations research (Virtual University of Pakistan)



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(1.)

Jathal

MTH-632

## Final term Solution

Q#01 prove that

imp

$$\frac{d}{dz} \sin z = \cos z$$

proof:

L.H.S

$$\frac{d}{dz} \sin z = \frac{d}{dz} \left( \frac{e^{iz} - e^{-iz}}{2i} \right)$$

$$= \frac{1}{2i} \left( \frac{d}{dz} e^{iz} - \frac{d}{dz} e^{-iz} \right)$$

$$= \frac{1}{2i} (ie^{iz} - (-ie^{-iz}))$$

$$= \frac{1}{2i} (ie^{iz} + ie^{-iz})$$

$$= \frac{1}{2i} (ie^{iz} + ie^{-iz})$$

hol

(2)

$$= \frac{e^{iz} + e^{-iz}}{2}$$

$$\therefore \frac{e^{iz} + e^{-iz}}{2} = \cos z$$

$$= \cos z$$

$$\therefore \frac{e^{iz} - e^{-iz}}{2i} = \sin z$$

$$\text{L.H.S} = \text{R.H.S}$$

Proved

Q#02 Show That

Imp

$$\frac{d}{dz} \cos z = -\sin z$$

Proof:

$$\frac{d}{dz} \cos z = \frac{d}{dz} \left( \frac{e^{iz} + e^{-iz}}{2} \right)$$

$$= \frac{1}{2} \left( \frac{d}{dz} e^{iz} + \frac{d}{dz} e^{-iz} \right)$$

$$= \frac{1}{2} (ie^{iz} + (-i)e^{-iz})$$

$$= \frac{i}{2} (e^{iz} - e^{-iz})$$

Multiply and div by i

$$= \frac{i}{2} \cdot \frac{i}{i} (e^{iz} - e^{-iz})$$



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$$= \frac{i^2}{2i} (e^{iz} - e^{-iz})$$

$$= (-1) \left( \frac{e^{iz} - e^{-iz}}{2i} \right)$$

$$\because i^2 = -1$$

$$= - \left( \frac{e^{iz} - e^{-iz}}{2i} \right)$$

$$= -\sin z$$

Proved

Show that

$$\cos z + i \sin z = e^{iz}$$

Proof:

We know That

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i} \quad \text{--- (i)}$$

$$\cos z = \frac{e^{iz} + e^{-iz}}{2} \quad \text{--- (ii)}$$

By adding eq (i) and (ii)

(4)

$$\cos z + i \sin z = \frac{e^{iz} + e^{-iz}}{2} + i \frac{e^{iz} - e^{-iz}}{2i}$$

$$= \frac{e^{iz} + e^{-iz}}{2} + \frac{e^{iz} - e^{-iz}}{2}$$

$$= e^{iz}$$

Show That

$$\sin(z_1 + z_2) = \sin z_1 \cos z_2 + \cos z_1 \sin z_2$$

Proof:

We know That

$$2 \sin z_1 \cos z_2 = \sin(z_1 + z_2) + \sin(z_1 - z_2) \quad \text{eq (1)}$$

Interchanging  $z_1$  by  $z_2$

$$2 \cos z_1 \sin z_2 = \sin(z_2 + z_1) + \sin(z_2 - z_1)$$

$$2 \cos z_1 \sin z_2 = \sin(z_1 + z_2) + \sin(-z_1 + z_2)$$

$$2 \cos z_1 \sin z_2 = \sin(z_1 + z_2) - \sin(z_1 - z_2) \quad \text{eq (2)}$$

By adding eq (1) and (2)



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$$\begin{aligned} 2 \sin z_1 \cos z_2 &= \sin(z_1 + z_2) + \sin(z_1 - z_2) \\ 2 \cos z_1 \sin z_2 &= \sin(z_1 + z_2) - \sin(z_1 - z_2) \\ + \end{aligned}$$

$$2(\sin z_1 \cos z_2 + \cos z_1 \sin z_2) = 2 \sin(z_1 + z_2)$$

OR

$$\sin(z_1 + z_2) = \sin z_1 \cos z_2 + \cos z_1 \sin z_2$$

Proved

Show That

$$\cos(z_1 + z_2) = \cos z_1 \cos z_2 - \sin z_1 \sin z_2$$

Proof:

As we know That

$$\sin(z_1 + z_2) = \sin z_1 \cos z_2 + \cos z_1 \sin z_2$$

By diff it w.r.t  $z_1$  we get

$$\cos(z_1 + z_2) = \cos z_1 \cos z_2 + (-\sin z_1) \sin z_2$$

$$\cos(z_1 + z_2) = \cos z_1 \cos z_2 - \sin z_1 \sin z_2$$

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Show That

$$\sin^2 z + \cos^2 z = 1$$

Proof:

We know That

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i} \quad \text{--- (i)}$$

$$\cos z = \frac{e^{iz} + e^{-iz}}{2} \quad \text{--- (ii)}$$

So L.H.S

$$\sin^2 z + \cos^2 z = \left( \frac{e^{iz} - e^{-iz}}{2i} \right)^2 + \left( \frac{e^{iz} + e^{-iz}}{2} \right)^2$$

$$= \frac{e^{2iz} + e^{-2iz} - 2e^{iz}e^{-iz}}{-4} + \frac{e^{2iz} + e^{-2iz} + 2e^{iz}e^{-iz}}{4}$$

$$= \frac{-e^{2iz} - e^{-2iz} + 2e^{iz}e^{-iz} + e^{2iz} + e^{-2iz} + 2e^{iz}e^{-iz}}{4}$$

$$= \frac{4e^{iz}e^{-iz}}{4}$$

$$= e^{iz}e^{-iz}$$

$$= e^{iz-iz}$$

$$= e^0$$

$$= 1$$

R.H.S



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Show that

$$1 + \tan^2 z = \sec^2 z$$

Proof:

From the identity

$$\sin^2 z + \cos^2 z = 1$$

We have

$$\frac{\sin^2 z}{\cos^2 z} + \frac{\cos^2 z}{\cos^2 z} = \frac{1}{\cos^2 z}$$

$$\tan^2 z + 1 = \sec^2 z$$

Proved

Show that

$$1 + \cot^2 z = \operatorname{cosec}^2 z$$

Proof:

= from the identity

$$\sin^2 z + \cos^2 z = 1$$

We have

$$\frac{\sin^2 z}{\sin^2 z} + \frac{\cos^2 z}{\sin^2 z} = \frac{1}{\sin^2 z}$$

$$1 + \cot^2 z = \operatorname{cosec}^2 z$$

Proved



(8)

Show That

$$|\sin z|^2 = \sin^2 x + \sinh^2 y$$

Solution:

As we know That

$$\sin z = \sin x \cosh y + i \cos x \sinh y$$

We find That

$$|\sin z|^2 = \sin^2 x \cosh^2 y + \cos^2 x \sinh^2 y$$

$$= \sin^2 x \cosh^2 y + \cos^2 x \sinh^2 y$$

$$= \sin^2 x (1 + \sinh^2 y) + (1 - \sin^2 x) \sinh^2 y$$

$$= \sin^2 x + \sin^2 x \sinh^2 y + \sinh^2 y - \sin^2 x \sinh^2 y$$

$$= \sin^2 x + \sinh^2 y$$

Proved

Jathol

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Show That

$$|\cos z|^2 = \cos^2 x + \sinh^2 y$$

Solution:

As we know That

$$\cos z = \cos x \cosh y + i \sin x \sinh y$$

from This we have

$$|\cos z|^2 = \cos^2 x \cosh^2 y + \sin^2 x \sinh^2 y$$

$$= \cos^2 x (1 + \sinh^2 y) + (1 - \cos^2 x) \sinh^2 y$$

$$= \cos^2 x + \cancel{\cos^2 x \sinh^2 y} + \sinh^2 y - \cancel{\cos^2 x \sinh^2 y}$$

$$= \cos^2 x + \sinh^2 y$$

provel

Show



(10)

R.H.S

$$\begin{aligned}
\cos(\bar{iz}) &= \cos(-ix/y) \\
&= \cos(-(y+ix)) \\
&= \cos(y+ix) \\
&= \cos y \cosh x -
\end{aligned}$$

Show That  $\underline{\text{v. Imp}}$

$$\overline{\cos(iz)} = \cos(\bar{iz})$$

Sol:

L.H.S

$$\overline{\cos(iz)} = \overline{\cos(-y+ix)}$$

$$= \overline{\cos(-y) \cosh x - i \sin(-y) \sinh x}$$

$$= \overline{\cos y \cosh x + i \sin y \sinh x}$$

$$= \cos y \cosh x - i \sin y \sinh x$$

R.H.S

$$\cos(\bar{iz}) = \cos(-y+ix)$$

$$= \cos(-y-ix)$$

$$= \cos(-(y+ix))$$

$$= \cos(y+ix)$$

$$\therefore iz = -y+ix$$

$$= \cos(-y)$$

$$= \cos y$$

(ii)

$$= \cos y \cosh u - \sin y \sinh u$$

$$L.H.S = R.H.S$$

Imp

Show that

$$\frac{d}{dz} \arcsin z = \frac{1}{\sqrt{1-z^2}}$$

Sol:

$$\text{let } w = \arcsin z$$

$$\therefore \arcsin z = \sin^{-1} z$$

$$z = \sin w$$

diff b.s w.r. to 'z'

$$\frac{d}{dz} z = \frac{d}{dz} \sin w$$

$$1 = \cos w \cdot \frac{dw}{dz}$$

$$\frac{1}{\cos w} = \frac{dw}{dz}$$

$$\frac{dw}{dz} = \frac{1}{\sqrt{1-\sin^2 w}}$$

$$\frac{dw}{dz}$$

$$1$$

$$\sqrt{1-z^2}$$



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$$\frac{d}{dz} w = \frac{1}{\sqrt{1-z^2}}$$

$$\Rightarrow \frac{d}{dz} \arcsin z = \frac{1}{\sqrt{1-z^2}}$$

Proved

Imp

Prove That

$$\int_0^{\pi+2i} \cos\left(\frac{z}{2}\right) dz = e + \frac{1}{e}$$

Solution:

L. H. S

$$\int_0^{\pi+2i} \cos\left(\frac{z}{2}\right) dz$$

$$= \left| \frac{\sin(z/2)}{1/2} \right|_0^{\pi+2i}$$

$$= 2 \left| \sin(z/2) \right|_0^{\pi+2i}$$

$$= 2 \left[ \sin\left(\frac{\pi+2i}{2}\right) - \sin(0) \right]$$

$$= 2 \left( \sin\left(\frac{\pi}{2} + i\right) - 0 \right)$$

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Since

formula:  $\sin z = \frac{e^{iz} - e^{-iz}}{2i}$

$$= \frac{1}{2i} \left( \frac{e^{i(\pi/2 + i)} - e^{-i(\pi/2 + i)}}{2i} \right)$$

$$= \frac{1}{i} \left( e^{\left(\frac{i\pi}{2} + i^2\right)} - e^{\left(-\frac{i\pi}{2} - i^2\right)} \right)$$

$$= \frac{1}{i} \left( e^{i\pi/2 - 1} - e^{-i\pi/2 + 1} \right)$$

$$= \frac{1}{i} \left( e^{i\pi/2} \cdot e^{-1} - e^{-i\pi/2} \cdot e^1 \right)$$

Since

formula:  $e^{iz} = \cos z + i \sin z$

$$= \frac{1}{i} \left( (\cos \pi/2 + i \sin \pi/2) e^{-1} \right.$$

$$\left. - (\cos(-\pi/2) + i \sin(-\pi/2)) e^1 \right)$$



(14)

$$= \frac{1}{i} ((0+i)e^{-1} - (0-i)e)$$

$$= \frac{1}{i} (ie^{-1} + ie)$$

$$= \frac{i}{i} (e + e^{-1})$$

$$= (e + 1/e)$$

$$= R.H.S$$

proved

Smp

find  $\int_C f(z) dz$  if

$$f(z) = \frac{z+2}{z}, \quad z = e^{i\theta}$$

Sol:

$$\int_C f(z) dz$$

$$= \int_0^\pi \left( \frac{z+2}{z} \right) dz$$

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$$= \int_0^{\pi} \left( \frac{z+2}{z} \right) (i2e^{i\theta}) d\theta$$

$$= \int_0^{\pi} \left( \frac{z}{z} + \frac{2}{z} \right) (i2e^{i\theta}) d\theta$$

$$= \int_0^{\pi} \left( 1 + \frac{2}{ze^{i\theta}} \right) (i2e^{i\theta}) d\theta$$

$$= \int_0^{\pi} \left( 1 + \frac{1}{e^{i\theta}} \right) (i2e^{i\theta}) d\theta$$

$$= \int_0^{\pi} \left( \frac{e^{i\theta} + 1}{e^{i\theta}} \right) (i2e^{i\theta}) d\theta$$

$$= 2i \int_0^{\pi} \frac{e^{i\theta} + 1}{e^{i\theta}} \cdot e^{i\theta} d\theta$$

$$= 2i \int_0^{\pi} (e^{i\theta} + 1) d\theta$$



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$$= 2i \left( \int_0^{\pi} e^{i\theta} d\theta + \int_0^{\pi} 1 d\theta \right)$$

$$= 2i \left[ \left| \frac{e^{i\theta}}{i} \right|_0^{\pi} + \left| \theta \right|_0^{\pi} \right]$$

$$= 2i \left[ \left( \frac{e^{i\pi}}{i} - \frac{e^0}{i} \right) + (\pi - 0) \right]$$

$$= 2i \left( \frac{e^{i\pi}}{i} - \frac{1}{i} \right) + 2i(\pi)$$

$$= \frac{2i}{i} (e^{i\pi} - 1) + 2i\pi$$

$$= 2((\cos \pi + i \sin \pi) - 1) + 2i\pi$$

$$= 2((-1 + 0) - 1) + 2i\pi$$

$$= 2(-1 - 1) + 2i\pi$$

$$= -4 + 2i\pi$$

$$\because e^{iz}$$

$$= \cos z + i \sin z$$

(17)

Imp

Find antiderivative of

$$\int_i^{i/2} e^{\pi z} dz$$

Solution:

$$\int_i^{i/2} e^{\pi z} dz$$

$$= \left| \frac{e^{\pi z}}{\pi} \right|_i^{i/2}$$

$$= \frac{1}{\pi} \left( e^{\pi i/2} - e^{i\pi} \right)$$

$$= \frac{1}{\pi} \left( e^{i\pi/2} - e^{i\pi} \right)$$

$$= \frac{1}{\pi} \left( \cos(\pi/2) + i \sin(\pi/2) - (\cos \pi + i \sin \pi) \right)$$

$$= \frac{1}{\pi} \left( (0 + i) - (-1 + 0) \right)$$

$$= \frac{1}{\pi} (i + 1)$$

$$= \frac{i + 1}{\pi}$$



Imp

(18)

Prove that  $U = x^3 - 3xy^2 + 3x^2y - 3y^2 + 1$

satisfies Laplace's equation.

Sol:

Given That

$$U = x^3 - 3xy^2 + 3x^2y - 3y^2 + 1$$

diff w.r.t 'x'

$$\frac{\partial U}{\partial x} = 3x^2 - 3y^2 + 6xy$$

$$\frac{\partial^2 U}{\partial x^2} = 6x + 6 \quad \text{--- (1)}$$

Now diff w.r.t 'y'

$$\frac{\partial U}{\partial y} = -6xy - 6y$$

$$\frac{\partial^2 U}{\partial y^2} = -6x - 6 \quad \text{--- (2)}$$

By adding eq (1) and (2)

$$\frac{\partial^2 U}{\partial x^2} + \frac{\partial^2 U}{\partial y^2} = 6x + 6 - 6x - 6$$

= 0

$\therefore U$  satisfies the Laplace equation.

Question: Imp  $Z_n = \frac{1}{n^3} - i$  solve it

for convergence also find value of  $\epsilon$ .

Solution:

$$Z_n = \frac{1}{n^3} - i$$

$$\lim_{n \rightarrow \infty} Z_n = \lim_{n \rightarrow \infty} \left( \frac{1}{n^3} - i \right)$$

$$= \lim_{n \rightarrow \infty} \left( \frac{1}{n^3} \right) - \lim_{n \rightarrow \infty} (i)$$

$$= \frac{1}{\infty} - i$$

$$= -i < 0 \quad (\text{converges})$$





Imp

(20)

Evaluate  $\bar{z}_1 - \bar{z}_2$  when  
 $z_1 = 5+i$  and  $z_2 = -8+3i$

Sol:

$$z_1 = 5+i, \quad z_2 = -8+3i$$

$$\bar{z}_1 = 5-i, \quad \bar{z}_2 = -8-3i$$

$$\bar{z}_1 - \bar{z}_2 = (5-i) - (-8-3i)$$

$$= 5-i+8+3i$$

$$= 2i+13$$

if  $x(t) = a \sin t \hat{e}_1 + a \cos t \hat{e}_2 + \dots$

Then solve  $x'(t)$

(21)

Imp

if  $x(t) = a \cos t \hat{e}_1 + a \sin t \hat{e}_2 + b t \hat{e}_3$   
then solve  $\frac{x'(t)}{|x'(t)|}$

Solution:

$$x(t) = a \cos t \hat{e}_1 + a \sin t \hat{e}_2 + b t \hat{e}_3$$

diff w.r.t "t"

$$x'(t) = -a \sin t \hat{e}_1 + a \cos t \hat{e}_2 + b \hat{e}_3$$

$$|x'(t)| = \sqrt{(-a \sin t)^2 + (a \cos t)^2 + b^2}$$

$$= \sqrt{a^2 \sin^2 t + a^2 \cos^2 t + b^2}$$

$$= \sqrt{a^2 (\sin^2 t + \cos^2 t) + b^2}$$

$$= \sqrt{a^2 + b^2}$$

So

$$\frac{x'(t)}{|x'(t)|} = \frac{-a \sin t \hat{e}_1 + a \cos t \hat{e}_2 + b \hat{e}_3}{\sqrt{a^2 + b^2}}$$



(22)

if  $\nabla f$  at  $P(1,1,0)$   
 $v = (1, 0, -3)$  and  $f = x^2 y z$

Solution:

$$f = x^2 y z$$

$$\frac{\partial f}{\partial x} = 2xyz, \quad \frac{\partial f}{\partial y} = x^2 z$$

$$\frac{\partial f}{\partial z} = x^2 y$$

at point  $P = (1, 1, 0)$

$$\frac{\partial f}{\partial x} = 0, \quad \frac{\partial f}{\partial y} = 0, \quad \frac{\partial f}{\partial z} = 1$$

Here

$$\nabla_p [f] = v \frac{\partial f}{\partial x}(P) + v \frac{\partial f}{\partial y}(P)$$

$$+ \frac{\partial f}{\partial z}(P)$$

$$= 1(0) + 0(0) + (-3)(1)$$

$$= -3$$

-3 is directional derivative.

(23)

Find Singularity if  
 $f(z) = \frac{1}{z^2 + 2z + 2}$

Solution:

Here  $z^2 + 2z + 2 = 0$

by quadratic formula

$$z = \frac{-2 \pm \sqrt{4 - 4(1)(2)}}{2}$$

$$= \frac{-2 \pm \sqrt{-4}}{2}$$

$$= \frac{-2 \pm 2i}{2}$$

$$= -1 \pm i$$

$z = -1 \pm i$  is singular point

Show that  $\sum_{n=1}^{\infty} \bar{z}_n = \bar{z}$

Solution:



(24)

$$Z_n = x_n + iy_n$$

$$\bar{Z}_n = x_n - iy_n$$

and

$$S = X + iY$$

$$\bar{S} = X - iY$$

$$\sum_{n=1}^{\infty} x_n = X \quad ; \quad \sum_{n=1}^{\infty} y_n = Y$$

So

$$\sum_{n=1}^{\infty} \bar{Z}_n = \sum_{n=1}^{\infty} (x_n - iy_n)$$

$$= \sum_{n=1}^{\infty} x_n - \sum_{n=1}^{\infty} iy_n$$

$$= X - iY$$

$$\sum_{n=1}^{\infty} \bar{Z}_n = \bar{S}$$

Proved

Show that  $(-1)^{1/\pi} = \exp(i)$

Solution:

L.H.S

(25)

$$= (-1)^{1/\pi}$$

$$= \exp \left[ \frac{1}{\pi} \log(-1) \right]$$

$$= \exp \left[ \frac{1}{\pi} \log(1) \right]$$

$$= \exp \left[ \frac{1}{\pi} \left( \ln|1| + i(\arg(1) + 2n\pi) \right) \right]$$

$$= \exp \left[ \frac{1}{\pi} (0 + (0 + \pi + 2n\pi)i) \right]$$

$$= \exp \left[ \frac{1}{\pi} \cdot \pi(1+2n)i \right]$$

$$= \exp[(1+2n)i]$$

$$= \exp(1+2n)i$$

$$= \text{R.H.S}$$

Prove That

$$d(fg) = gdf + fdg$$

Solution:

$$d(fg) = \sum \frac{\partial}{\partial x_i} (fg) dx_i$$

$$= \sum \left( \frac{\partial f}{\partial x_i} g + f \frac{\partial g}{\partial x_i} \right) dx_i$$



(26)

$$= \sum \frac{\partial f}{\partial n_i} g \, dn_i + \sum \frac{\partial g}{\partial n_i} f \, dn_i$$

$$= g \sum \frac{\partial f}{\partial n_i} \, dn_i + f \sum \frac{\partial g}{\partial n_i} \, dn_i$$

$$= g \, df + f \, dg$$

R.H.S

Show That the function  
 $f(z) = \bar{z}$  nowhere diff.

Solution:

Given That

$$u + iv = x - iy$$

$$u = x, \quad v = -y$$

$$u_x = 1, \quad v_x = 0$$

$$u_y = 0, \quad v_y = -1$$

here  $u_x \neq v_y$

So  $f(z) = \bar{z}$  nowhere  
differentiable.

(27)

Test the analyticity of function  $w = \sin z$

Sol:

$$\text{let } w = f(z)$$

$$f(z) = \sin z$$

$$u + iv = \sin(x + iy)$$

$$= \sin x \cosh y + i \cos x \sinh y$$

Now

$$u = \sin x \cosh y \quad ; \quad v = \cos x \sinh y$$

$$u_x = \cos x \cosh y \quad ; \quad v_x = -\sin x \sinh y$$

$$u_y = \sin x \sinh y \quad ; \quad v_y = \cos x \cosh y$$

$$u_x = v_y \quad \text{and} \quad u_y = -v_x$$

$\therefore$  function is analytic.

Show that  $f(z) = e^z$  is analytic

Sol:  $f(z) = e^z$

$$u + iv = e^{x+iy}$$

$$= e^x \cdot e^{iy}$$

$$= e^x (\cos y + i \sin y)$$



(28).

$$u+iv = e^x \cos y + i e^x \sin y$$

$$u = e^x \cos y, \quad v = e^x \sin y$$

$$u_x = e^x \cos y, \quad v_x = e^x \sin y$$

$$u_y = -e^x \sin y, \quad v_y = e^x \cos y$$

here

$$u_x = v_y \quad \text{and} \quad u_y = -v_x$$

So function is analytic.

Given an example such that  $u$  and  $v$  are harmonic but  $u+iv$  is not analytic.  
Solution.

consider the function

$$f(z) = z$$

$$u+iv = x-iy$$

$$u = x, \quad v = -y$$

$$u_x = 1, \quad v_x = 0$$

$$u_{xx} = 0, \quad v_{xx} = 0$$

$$u_y = 0, \quad v_y = -1$$

$$u_{yy} = 0, \quad v_{yy} = 0$$

here Cauchy-Riemann equation not

$$u_x \neq v_y$$

satisfied.

(29.)

$\therefore f(z)$  is not analytic

But  $U_{xx} + V_{yy} = 0$  and  $V_{xx} + U_{yy} = 0$   
gives  $U$  and  $V$  are harmonic.

Write  $-3i^4$  in the form of  
 $a+ib$   
Soln:

$$\text{let } z = -3i^4$$

$$z = -3(i^4)$$

$$z = -3(-1)^2$$

$$= -3$$

$$z = -3 + 0i$$

Find the arc length of

$$s = \int_a^b \left| \frac{dr}{dt} \right| dt$$

$$r(t) = (\cos t, \sin t, t)$$

$$0 \leq t \leq 2\pi$$

Solution:



(30)

We start from The  
are length

$$S(t) = \int_a^b ||x'(t)|| dt$$

here

$$x(t) = (\cos t, \sin t, t)$$

$$x'(t) = (-\sin t, \cos t, 1)$$

$$||x'(t)|| = \sqrt{\sin^2 t + \cos^2 t + 1}$$

$$= \sqrt{1+1}$$

$$= \sqrt{2}$$

So

$$\int_0^{2\pi} \sqrt{2} dt$$

$$\sqrt{2} \int_0^{2\pi} dt$$

$$\sqrt{2} \left| t \right|_0^{2\pi}$$

(31)

$$= \sqrt{2} (2\pi - 0)$$

$$S(t) = 2\pi\sqrt{2}$$

made by  
Tathat



$$\int_0^{\pi} \frac{\cosh}{z^4} dz \quad \text{by Cauchy}$$

integral formula.

Solution:

We know the Cauchy Integral formula:

$$\int_C \frac{f(z)}{(z-z_0)^{n+1}} dz = \frac{2\pi}{n!} \left| f^{(n)}(z_0) \right|$$



(32)

$$\int_C \frac{\cosh z}{z^4} dz$$

$$= \int_C \frac{\cosh z}{(z - z_0)^{3+1}} dz$$

$$= \frac{2\pi}{3!} \left| \frac{d^3}{dz^3} \cosh z \right|_{z_0=0}$$

$$= \frac{2\pi}{3!} (0)$$

$$= 0$$

$U(x, y) = 2x(1-y)$  harmonic  
in some domain and find a  
harmonic conjugate  $V(x, y)$   
when

$$U(x, y) = 2x(1-y)$$

Solution:

It is straightforward  
to show that  $U_{xx} + U_{yy} = 0$   
When  $U(x, y) = 2x(1-y)$   
to find harmonic conjugation

(33)

$v(x, y)$ , we start with

$$u_x = 2 - 2y \quad \text{Now}$$

$$u_x = v_y \Rightarrow v_y = 2 - 2y$$

$$\Rightarrow v(x, y) = \underbrace{2y - y^2}_{v_y} + \underbrace{\phi(x)}_{v_x}$$

then

$$u_y = -v_x \Rightarrow -2x = -\phi'(x) \Rightarrow \phi'(x) = 2x$$

$$\Rightarrow \phi(x) = x^2 + C$$

consequently

$$\begin{aligned} v(x, y) &= 2y - y^2 + (x^2 + C) \\ &= x^2 - y^2 + 2y + C \end{aligned}$$

$$\int_C \frac{e^{-z}}{(z - \pi i/2)} dz \quad \text{by Cauchy}$$

integral.

solution:

$$\int_C \frac{e^{-z}}{(z - \pi i/2)^{0+1}}$$

Since  $n=1$

$$\therefore f(z) = e^{-z}$$

$$f'(z) = e^{-z}$$

$$f'(\pi i/2) = e^{-\pi i/2}$$

$$\frac{2\pi i}{1!} \left| f'(\pi i/2) \right|$$



(34)

$$= 2\pi i \left( e^{-i\frac{\pi}{2}} \right)$$

$$= 2\pi i \left( \cos\left(-\frac{\pi}{2}\right) + i \sin\left(-\frac{\pi}{2}\right) \right)$$

$$= 2\pi i (0 - i)$$

$$= -2\pi i^2$$

$$= 2\pi$$

Find analytical region

$$f(z) = (x-y)^2 + 2i(x+y)$$

Sol:

$$u + iv = (x-y)^2 + 2i(x+y)$$

$$u = (x-y)^2, \quad v = 2(x+y)$$

$$, \quad u_x = 2$$

$$u_x = 2(x-y), \quad v_y = 2$$

$$u_y = 2(x-y)(-1),$$

$$\Rightarrow -2(x-y),$$

$\therefore f(z)$  is analytic, So

$u$  and  $v$  should satisfy

Cauchy-equation.

$$u_x = v_y \Rightarrow 2(x-y) = 2$$

$$\Rightarrow x-y = 1$$

(35)

also

$$u_y = -v_x \Rightarrow -2(n-y) = -2 \\ \Rightarrow n-y = 1$$

$\Rightarrow f(z)$  is analytic at  
all points satisfying  
 $n+y=1$ .

hence the analytic  
region is The line.

if  $f(z) = z(2-z)$  then  
find  $f(1+i)$

Solution:  $f(z) = 2z - z^2$

$$f(1+i) = 2(1+i) - (1+i)^2$$

$$= 2 + 2i - 1 - i^2 - 2i$$

$$= 2 - 1 + 1$$

$$= 2$$



(36)

Serret Frenet formulas

Proof:  $\hat{t}, \hat{n}, \hat{b}$

$$i) \frac{d\hat{t}}{ds} = \kappa \hat{n} \Rightarrow \hat{t}' = \kappa \hat{n}$$

$$ii) \frac{d\hat{r}}{ds} = \tau \hat{b} \Rightarrow \hat{n}' = \tau \hat{b} - \kappa \hat{t}$$

$$iii) \frac{d\hat{b}}{ds} = -\tau \hat{n} \Rightarrow \hat{b}' = -\tau \hat{n}$$

Proof of (i)

$$\hat{t}' = \kappa \hat{n}$$

Proof:

$\hat{t}$  is unit vector

$$\hat{t} \cdot \hat{t} = 1 \quad \text{--- (i)}$$

diff w.r.t 's'

$$\frac{d\hat{t}}{ds} \cdot \hat{t} + \hat{t} \cdot \frac{d\hat{t}}{ds} = 0$$

$$2\hat{t} \cdot \hat{t}' = 0$$

$$\hat{t} \cdot \hat{t}' = 0 \quad \text{--- (ii)}$$

$\hat{t}'$  is  $\perp$  to  $\hat{t}$

(37)

$$\frac{d\hat{t}}{ds} = \hat{t}' \quad \text{along } \hat{n}$$

$$\frac{d\hat{t}}{ds} = \pm \left| \frac{d\hat{t}}{ds} \right| \hat{n}$$

$$\hat{t}' = \kappa \hat{n}$$

Proved

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